

Task 1

① $\mathcal{B} = \{(1,0,0), (0,1,0), (0,0,1)\}$

$$T(1,0,0) = (1,0)$$

$$T(0,1,0) = (-1,0)$$

$$T(0,0,1) = (0,2)$$

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

② $T(1,0,0,0,0,0) = (2,0,0,0)$

$$T(0,1,0,0,0,0) = (-1,1,0,0)$$

$$T(0,0,1,0,0,0) = (0,1,0,0)$$

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

③ $T(1,0) = (2,3,1)$

$$T(0,1) = (-1,4,0)$$

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 2 & -1 \\ 3 & 4 \\ 1 & 0 \end{bmatrix}$$

④ $\mathcal{B} = \text{standard ordered bases for } \mathbb{R}^n, \mathcal{B} = \{\beta_1, \beta_2, \beta_3, \dots, \beta_n\}$

$$T(\beta_1) = \beta_n$$

$$T(\beta_2) = \beta_{n-1}$$

$$[T]_{\mathcal{B}} = \delta_{i, n+1-j} = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & \dots & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 \end{bmatrix}$$

Task 2

$$T(a_1, a_2) = (a_1 - a_2, a_1, 2a_1 + a_2)$$

$$T(1, 0) = (1, 1, 2)$$

$$T(0, 1) = (-1, 0, 1)$$

$$[T]_{\beta} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$$\gamma = \{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$$

$$(1, 1, 2) = b(1, 1, 0) + c(0, 1, 1) + d(2, 2, 3) \quad b, c, d \in \mathbb{R}$$

$$\begin{cases} b + 2d = 1 \\ b + c + 2d = 1 \\ c + 3d = 2 \end{cases} \quad \begin{cases} b = -\frac{1}{3} \\ c = 0 \\ d = \frac{2}{3} \end{cases}$$

$$(-1, 0, 1) = e(1, 1, 0) + f(0, 1, 1) + g(2, 2, 3) \quad e, f, g \in \mathbb{R}$$

$$\begin{cases} e + 2g = 1 \\ e + f + 2g = 0 \\ d + 3g = 1 \end{cases} \quad \begin{cases} e = 1 \\ f = 1 \\ g = 0 \end{cases}$$

$$[T]_{\beta}^{\gamma} = \begin{bmatrix} -\frac{1}{3} & 1 \\ \frac{2}{3} & 1 \\ \frac{1}{3} & 0 \end{bmatrix}$$

Task 2

$$\alpha = \{(1, 2), (2, 3)\}$$

$$T(1, 2) = (-1, 1, 4)$$

$$T(2, 3) = (-1, 2, 7)$$

$$\beta = \{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$$

$$(-1, 1, 4) = a(1, 1, 0) + b(0, 1, 1) + c(2, 2, 3) \quad a, b, c \in \mathbb{R}$$

$$\begin{cases} a + 2c = -1 \\ a + b + 2c = 1 \\ b + 3c = 4 \end{cases} \quad \begin{cases} a = -\frac{7}{3} \\ b = 2 \\ c = \frac{2}{3} \end{cases}$$

$$(-1, 2, 7) = a(1, 1, 0) + b(0, 1, 1) + c(2, 2, 3) \quad a, b, c \in \mathbb{R}$$

$$\begin{cases} a + 2c = -1 \\ a + b + 2c = 2 \\ b + 3c = 7 \end{cases} \quad \begin{cases} a = -\frac{11}{3} \\ b = 3 \\ c = \frac{4}{3} \end{cases}$$

$$[T]_{\alpha}^{\beta} = \begin{bmatrix} -\frac{7}{3} & -\frac{11}{3} \\ 2 & 3 \\ \frac{2}{3} & \frac{4}{3} \end{bmatrix}$$

Task 3

$$\text{tr}(A) = \sum_{i=1}^n A_{ii}$$

Prove $\text{tr}(AB) = \text{tr}(BA)$

Let $C = AB$, $C' = BA$

$$C_{ii} = \sum_{j=1}^n (A_{ij} \times B_{ji}) \quad \text{tr}(AB) = \text{tr}(C) = \sum_{i=1}^n C_{ii} = \sum_{i=1}^n \left(\sum_{j=1}^n (A_{ij} \times B_{ji}) \right)$$

$$C'_{ii} = \sum_{j=1}^n (B_{ij} \times A_{ji}) \quad \text{tr}(BA) = \text{tr}(C') = \sum_{i=1}^n C'_{ii} = \sum_{i=1}^n \left(\sum_{j=1}^n (B_{ij} \times A_{ji}) \right)$$

$$\sum_{i=1}^n \left(\sum_{j=1}^n (A_{ij} \times B_{ji}) \right) = \sum_{i=1}^n \sum_{j=1}^n (A_{ij} \times B_{ji}) = \sum_{j=1}^n \sum_{i=1}^n (B_{ji} \times A_{ij}) = \sum_{j=1}^n C'_{jj}$$

$$\text{tr}(AB) = \text{tr}(C) = \sum_{i=1}^n C_{ii} = \sum_{j=1}^n C'_{jj} = \text{tr}(C') = \text{tr}(BA)$$

Task 4

①

$$T(A) = A^t, \quad A \in M_{2 \times 2}(\mathbb{R})$$

$$T\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$[T]_{\alpha} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[T(A)]_{\alpha} = [T]_{\alpha} [A]_{\alpha} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} [A]_{\alpha}$$

$$[A]_{\alpha} = \begin{bmatrix} 1 \\ 4 \\ -1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 4 \\ 2 \end{bmatrix}$$

$$[T(A)]_{\alpha} = \begin{bmatrix} 1 \\ -1 \\ 4 \\ 2 \end{bmatrix}$$

②

$$T(f(x)) = f(2)$$

$$T(1) = 1$$

$$T(x) = 2$$

$$T(x^2) = 4$$

$$[T]_{\beta}^{\gamma} = [1 \ 2 \ 4]$$

$$[T(f(x))]_{\beta}^{\gamma} = [T]_{\beta}^{\gamma} \times [f(x)]_{\beta} = [1 \ 2 \ 4] \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} = [13]$$